

HUMA, China

WMO: 503530

Lat: 51.7261N

Lon: 126.6389E

Elev: 176

StdP: 99.23

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.5	-35.4	-40.6	0.1	-37.2	-38.4	0.1	-35.1	8.5	-18.7	7.6	-17.8	0.3	180	0.578

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	31.3	20.0	29.5	19.7	27.7	19.0	22.6	28.1	21.7	26.7	20.7	25.3	3.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.8	15.8	25.0	20.0	15.0	24.2	19.1	14.2	23.1	67.7	28.3	64.3	26.7	60.9	25.5	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.7	7.3	6.1	-39.4	35.1	2.8	2.0	-41.4	36.5	-43.0	37.6	-44.6	38.7	-46.7	40.2		
			-39.4	24.0	2.9	1.0	-41.5	24.8	-43.2	25.4	-44.8	25.9	-47.0	26.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.1	-24.3	-19.0	-8.8	3.2	11.7	18.6	21.4	18.8	11.6	1.4	-13.1	-23.6	(6)
(7)		DBStd	17.05	5.54	5.33	6.24	4.52	4.16	3.57	2.88	3.21	4.30	5.00	7.11	5.43	(7)
(8)		HDD10.0	4733	1062	813	584	209	28	1	0	1	32	270	693	1041	(8)
(9)		HDD18.3	6921	1321	1046	842	453	209	38	7	33	203	526	943	1299	(9)
(10)		CDD10.0	1054	0	0	0	6	82	259	352	272	81	3	0	0	(10)
(11)		CDD18.3	200	0	0	0	0	4	47	101	46	2	0	0	0	(11)
(12)		CDH23.3	1993	0	0	0	8	111	571	857	402	45	1	0	0	(12)
(13)		CDH26.7	609	0	0	0	1	27	196	280	101	4	0	0	0	(13)
(14)	Wind	WSAvg	2.0	1.4	1.6	2.4	3.0	2.8	2.1	2.0	1.8	2.0	2.3	1.7	1.3	(14)
(15)	Precipitation	PrecAvg	467	4	3	9	27	39	71	115	110	55	20	11	7	(15)
(16)		PrecMax	615	13	11	25	95	84	172	189	249	185	47	45	17	(16)
(17)		PrecMin	337	0	0	1	3	11	15	45	43	12	4	2	1	(17)
(18)		PrecStd	75	3	3	6	19	22	40	40	48	36	13	9	4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-8.1	-1.2	10.8	23.9	30.2	34.4	34.7	31.7	27.0	19.6	5.4	-5.4	(19)
(20)			MCWB	-9.6	-4.2	3.5	10.3	15.1	19.4	22.3	20.5	17.7	10.3	1.2	-7.2	(20)
(21)		2%	DB	-11.2	-4.9	6.9	18.3	26.5	31.0	31.5	29.3	24.6	15.1	1.8	-9.6	(21)
(22)			MCWB	-12.5	-6.8	1.1	8.0	13.6	18.7	21.2	21.0	16.4	8.1	-1.0	-10.8	(22)
(23)		5%	DB	-13.5	-7.8	3.8	14.9	23.7	28.9	29.6	27.4	22.1	12.2	-0.6	-13.2	(23)
(24)			MCWB	-14.5	-9.5	-0.6	6.4	12.3	18.1	20.7	20.0	15.1	6.3	-2.4	-14.1	(24)
(25)		10%	DB	-15.8	-10.3	1.3	12.2	21.0	26.6	27.8	25.5	19.7	9.4	-2.9	-15.5	(25)
(26)			MCWB	-16.6	-11.6	-2.0	4.7	11.2	17.3	20.3	19.1	13.7	4.5	-4.6	-16.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-9.8	-4.1	4.0	11.1	16.4	22.3	23.9	23.3	19.4	11.6	1.6	-7.0	(27)
(28)			MCD B	-8.3	-1.7	10.1	22.3	26.5	30.1	29.9	28.6	23.9	18.3	4.8	-5.1	(28)
(29)		2%	WB	-12.5	-6.7	1.5	8.5	14.9	20.5	22.9	22.1	17.6	8.6	-0.5	-10.8	(29)
(30)			MCD B	-11.4	-4.8	6.1	17.2	23.2	27.0	28.7	26.8	22.4	14.0	1.5	-9.5	(30)
(31)		5%	WB	-14.4	-9.4	-0.3	6.8	13.5	19.4	22.1	21.2	16.2	6.7	-2.4	-14.1	(31)
(32)			MCD B	-13.4	-7.8	3.6	14.1	21.2	25.8	27.5	25.6	20.6	11.2	-0.6	-13.2	(32)
(33)		10%	WB	-16.6	-11.5	-1.9	5.1	12.2	18.4	21.3	20.1	14.7	4.9	-4.6	-16.3	(33)
(34)			MCD B	-15.8	-10.3	1.3	11.2	19.0	24.8	26.2	24.1	18.8	9.1	-2.9	-15.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.8	13.8	13.3	12.3	13.0	12.3	10.4	11.1	12.4	10.9	10.3	9.6	(35)
(36)			MCD BR	11.7	15.8	14.8	18.1	18.7	16.4	14.0	14.4	15.8	15.8	11.7	10.6	(36)
(37)			MCWBR	10.6	13.8	10.0	10.2	8.3	6.4	5.5	6.8	8.1	9.4	9.1	9.7	(37)
(38)		5% WB	MCD BR	11.5	15.7	14.2	16.5	15.3	13.4	11.3	11.8	13.6	13.9	11.3	10.4	(38)
(39)			MCWBR	10.5	13.7	9.7	9.6	7.2	6.1	5.3	6.2	7.6	8.8	8.8	9.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.265	0.298	0.375	0.541	0.613	0.515	0.481	0.421	0.385	0.371	0.310	0.272	(40)	
(41)		taud	2.160	2.089	1.934	1.699	1.633	1.963	2.119	2.279	2.301	2.179	2.153	2.107	(41)	
(42)		Ebn at Noon	733	805	793	693	666	752	773	805	792	707	655	640	(42)	
(43)		Edh at Noon	71	103	148	214	240	174	147	119	103	95	72	64	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.49	2.68	4.35	5.12	5.39	6.02	5.43	4.76	3.75	2.46	1.55	1.15	(44)	
(45)		RadStd	0.05	0.07	0.20	0.54	0.42	0.67	0.48	0.32	0.29	0.15	0.07	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page